Rerandomization simulation report

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Process for rerandomization simulation

To generate simulation data I did the following for each outcome j in our set of outcomes across our empirical dataset (excepting a few studies that were inaccessible):

1) Load the empirical data for outcome j, patch missing data, etc.
2) Calculate the empirical estimates of the treatment effects for outcome j, and the associated empirical standard errors.
3) Repeatedly, for simulation runs $r = 1, \ldots, R$, permute the treatment assignment across clusters and within blocks. I then re-estimated the impacts given the permuted assignment.
4) I then, within each run and outcome, calculated the range of estimates, and the ratio of the largest to smallest standard error, across all the estimators that were defined for that run. I did this both overall, and for the group of estimators within each estimand (Person or Cluster).
5) I finally saved the set of results to a file.

Due to the permutation, there is no actual treatment effect. Any actual treatment effect is pushed into individual and cluster variation. For example, if a cluster had a positive average effect, the permutation simulation views this cluster as just having a slightly higher intercept than it would have had, had it been a control cluster in reality. The structure of the data, the relationship of covariates to outcome, etc., is all preserved by this approach.

The datasets missing from this analysis

We didn't run the simulation on all the empirical datasets due to privacy restrictions and computational feasibility. The datasets dropped are the following:

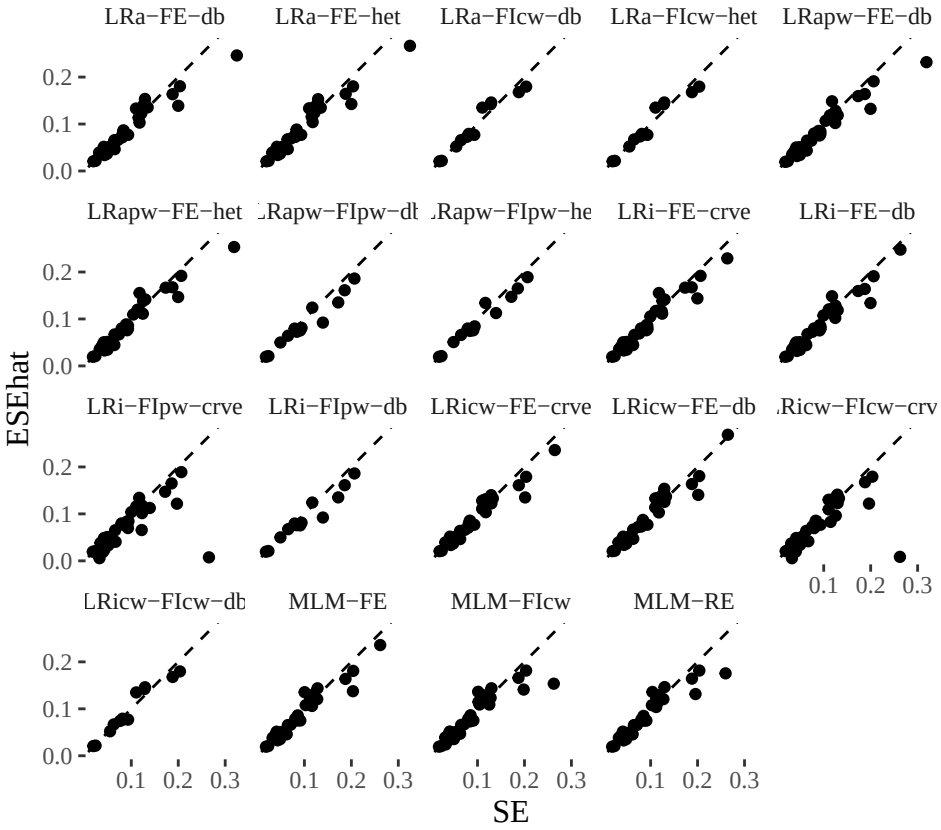
```
##
##
## Table: Dropped Datasets
##
## |UniqueID                 |      n|   J|
## |:------------------------|------:|---:|
## |CARES_DC_PT_Incred       |   1298|  52|
## |CARES_TR_BPI_Incred      |   1318|  52|
## |CARES_TR_WRS_Incred      |   1320|  52|
## |CARES_DC_EMIT_PTH        |   1229|  52|
## |CARES_DC_CST_COMPTOT_PTH |   1206|  52|
## |CARES_TR_SSRS_PTH        |   1276|  52|
## |CARES_DC_PT_Tools_Mind   |   1268|  52|
## |CARES_TR_WRS_Tools_Mind  |   1293|  52|
## |CLC_GRCCSS_POST          |   9557|  28|
## |CLC_PCCRERCORE           |  16465|  28|
## |MathPD_MTSS              |   2132|  39|
## |MathPD_MTSS_FRACTDEC     |   2132|  39|
```

```
## |MathPD_MTSS_PCTRATIO     |   2132|  39|
## |jg_crt_6                 |  15993| 802|
## |jg_crt_52                | 102869| 508|
```

How do the simulation estimated SEs compare to the actual simulation SE?

Within the simulation, we can compare the average estimated SE to the standard deviation of point estimates across the permutations. We include the actual empirical data as one of the "simulation permutations."

```
## Warning: Removed 132 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



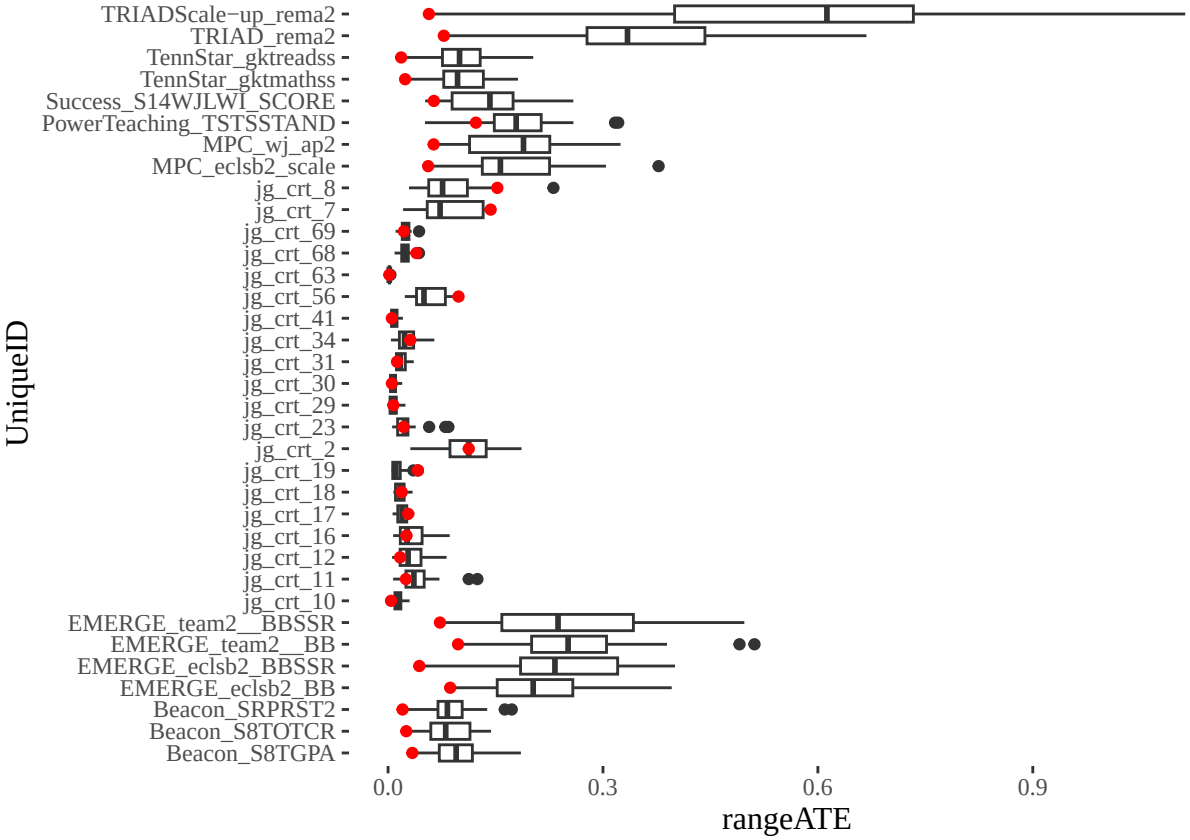
Some methods do not have standard errors, so they are not in the above plot. Counts of the methods with and without standard errors are below.

method	n	SE_def	ESEhat_def
LRa-FE-db	35	35	35
LRa-FE-het	35	35	35
LRapw-FE-db	35	35	35
LRapw-FE-het	35	35	35
LRi-FE-crve	35	35	35
LRi-FE-db	35	35	35
LRi-FIpw-crve	35	35	35
LRicw-FE-crve	35	35	35
LRicw-FE-db	35	35	35
LRicw-FIcw-crve	35	35	35

method	n	SE_def	ESEhat_def
MLM-FE	35	35	35
MLM-FIcw	35	35	35
MLM-RE	35	35	35
LRa-FIcw-db	35	35	13
LRa-FIcw-het	35	35	13
LRapw-FIpw-db	35	35	13
LRapw-FIpw-het	35	35	13
LRi-FIpw-db	35	35	13
LRicw-FIcw-db	35	35	13

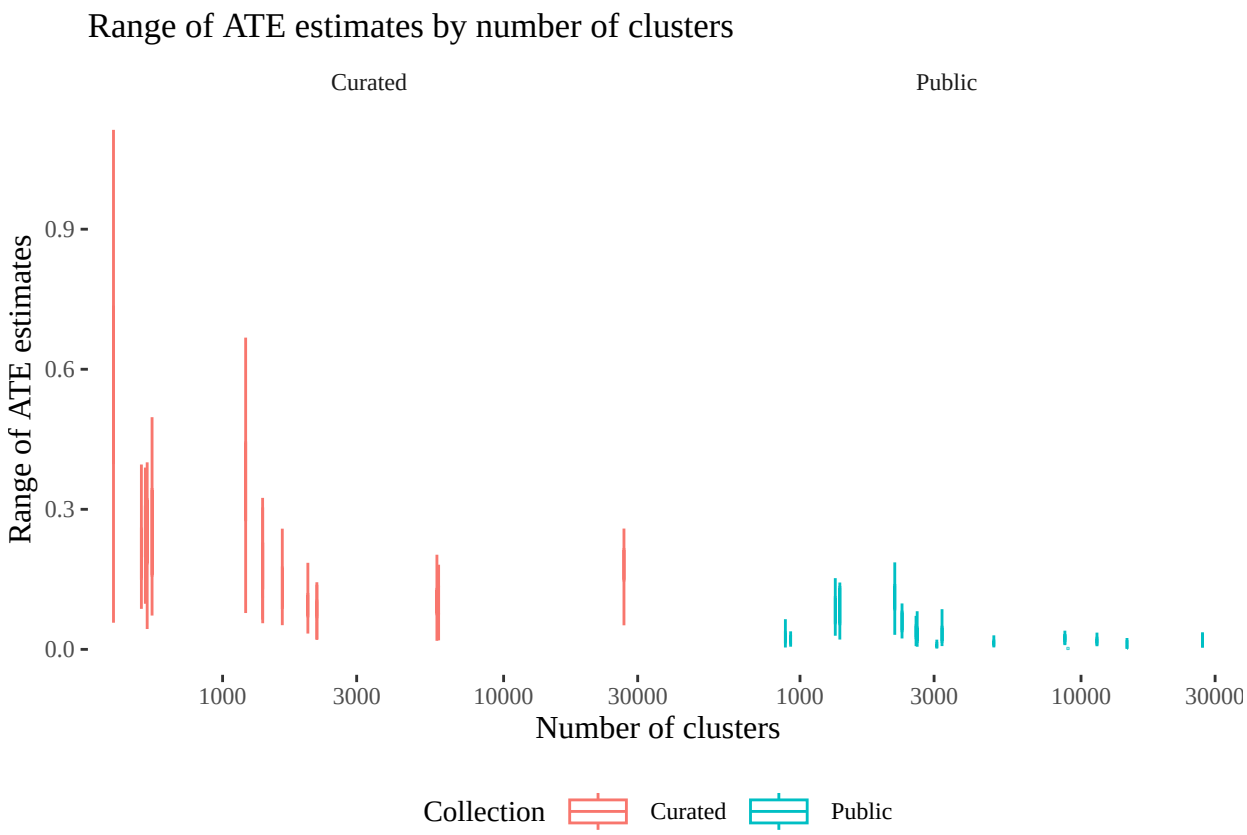
The actual vs. permutation estimates

For each study, we can look at the actual ATE range from the empirical data (red dot) vs. the ranges we get from the permutations (the boxplot).

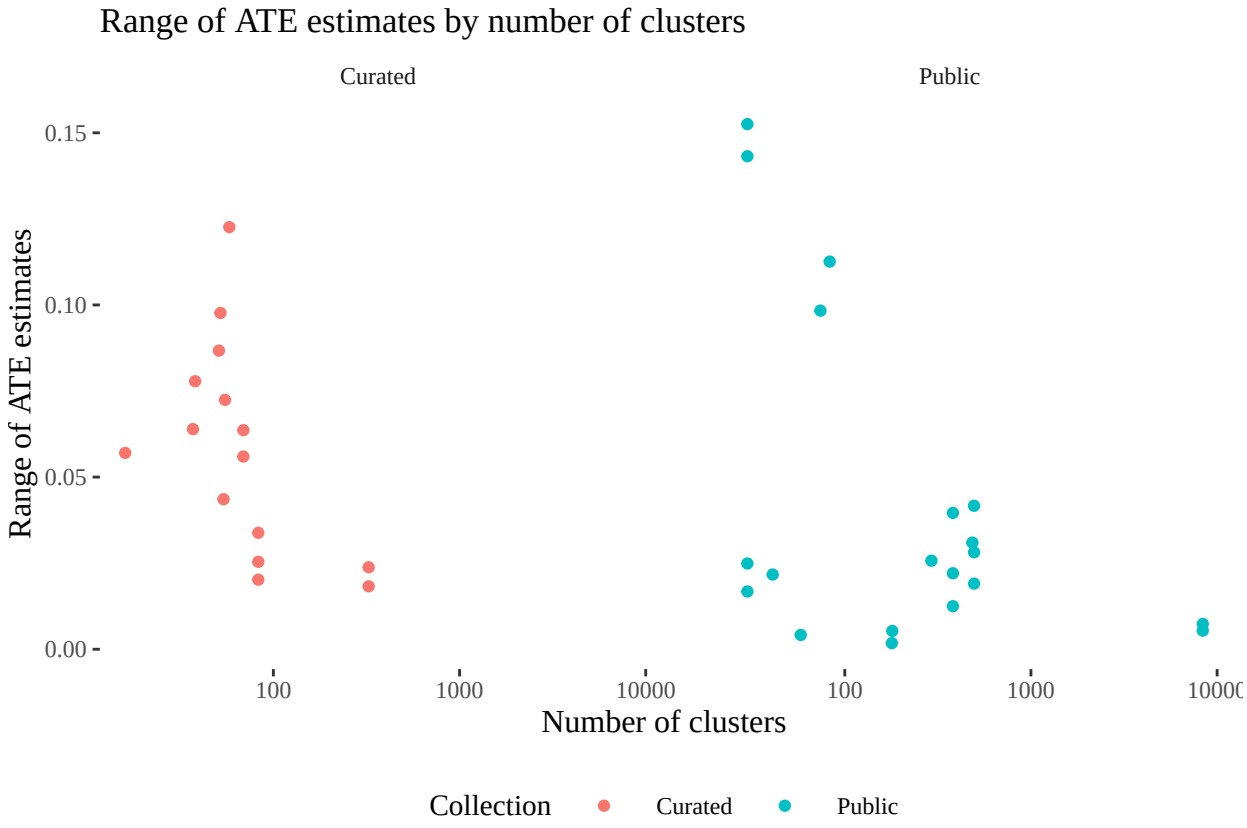


Weirdly, for the curated data, all the actual ranges are quite low. We also see notable differences between the two sets of data, with curated having much wider ranges in general.

We can look at the range of ATE estimates vs size of study (number of clusters):

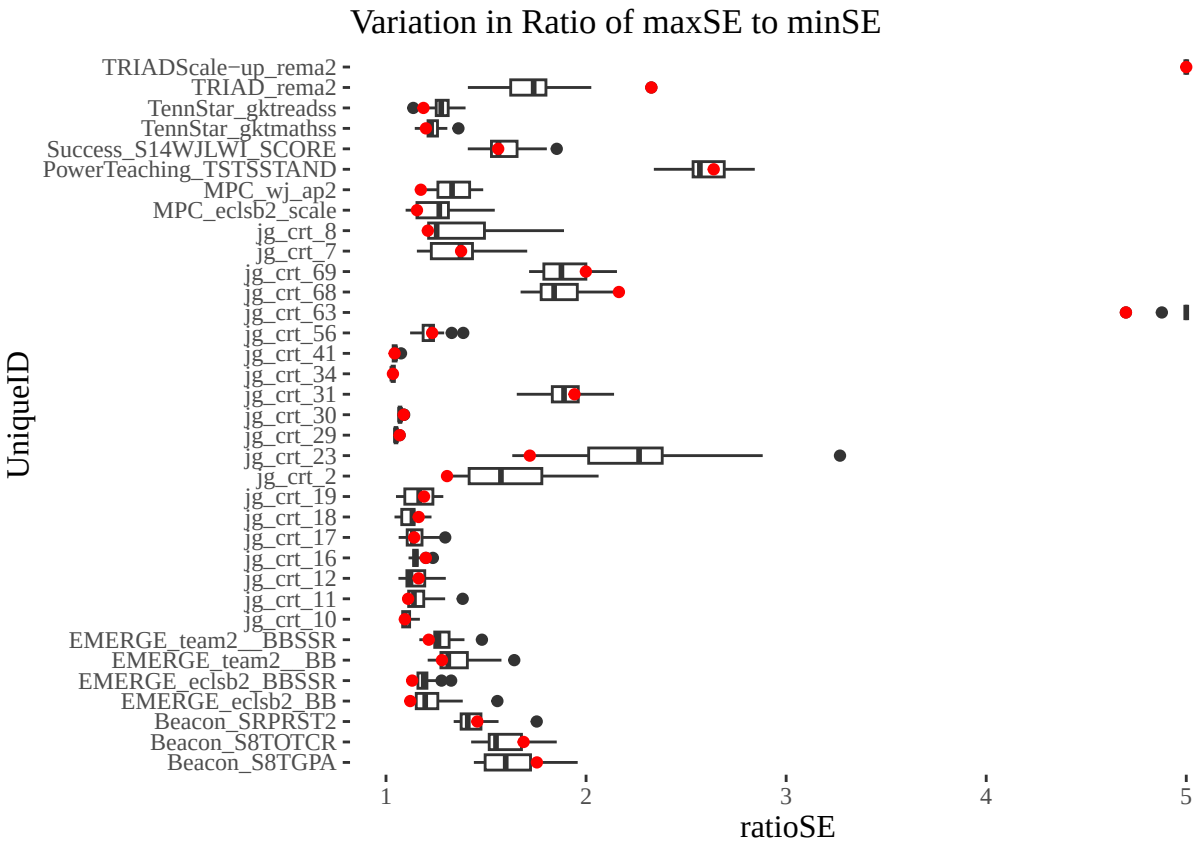


For reference, here are the set of empirical results:

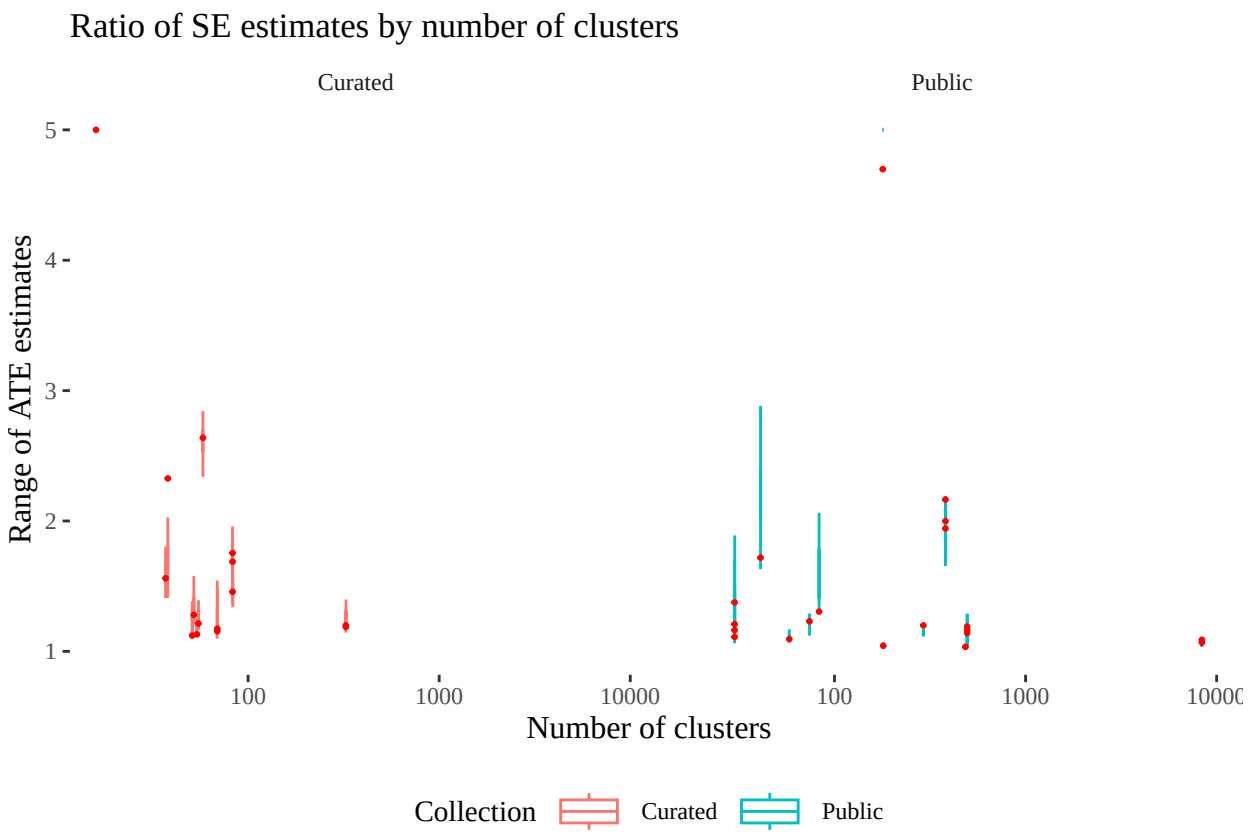


Variation in Standard Errors

We next look at how much the ratio of the max to min estimated standard errors vary across the permutations



We again look by number of clusters (the dots are the empirical values):



Imposter Plots

We can look at all the runs for all the datasets, and then for each run id r, make a histogram of the ranges across all the datasets, mimicking our core figures from the empirical paper. We then insert the real estimated values as an "imposter" so we could see if the estimated distributions looked like the actual one.

Aggregate statistics

For each run, we can look at the mean and standard deviation of the ranges of estimates across the datasets in that run.

Commentary